Test 8:

READING PASSAGE 1

You should spend about 20 minutes on Questions 1–13, which are based on Reading Passage 1 below.

Rubber

T. and W. Musgrove discuss the origins and early uses of rubber

The plants that produce rubber are spread right across the globe, and grow in many different habitats. One might think it likely, therefore, that humankind has known about rubber for thousands of years. Yet, unlike other crops of economic importance, rubber led a relatively anonymous life until the last 150 years or so. The Indians of South America appear to be the first people to have understood the properties of rubber, and the Aztecs of what is now Mexico were the first to be recorded using the substance; a wall painting dating back to the sixth century depicts a scene of a tribute offering of crude rubber.

With the arrival of Columbus in the Americas and the resulting Spanish influx, further evidence starts to appear concerning the Native American use of rubber. Antonio de Herrera y Tordesillas describes a ritual game played with a rubber ball at the court of the Aztec Emperor Montezuma II, and the Mayan and Toltec people are known to have taken part in similar activities. Rubber was also used to make raincoats, shoes, jars, torches and musical instruments, all of which must have been made from the indigenous *Castilla elastica*, as the Para rubber plant now favoured for rubber cultivation does not grow in the Mexican region.

The first description of latex (liquid rubber) extraction was made by Juan de Torquemada, who noted that if a receptacle was not at hand the Native Americans would place the latex on their bodies to allow it to solidify. However, no real interest in rubber was shown by any European until Charles de la Condamine, a French mathematician, published an account of his journey to South America in 1735. The journey was undertaken on behalf of the Paris Academy of Sciences to measure an arc of the meridian line on the equator, but the journey home was to turn out to be more significant than the true purpose of the trip. Condamine explored Brazil and Peru and discovered how the local people used one single piece of coagulated latex to make boots. The boots were impervious to water and, when smoked, looked like real leather.

In 1747 the first description of the rubber tree and latex tapping was made by a military engineer and amateur botanist, François Fresneau, who was posted to French Guiana. The publications of Condamine and Fresneau created considerable excitement among French scientists, and an attempt was made to discover a solvent that could turn the crude rubber into a substance for commercial exploitation.

In 1818 a British medical student named James Syme first used rubber to make waterproof cloth. Another early use of the substance was as an eraser of pencil marks, hence the name ‘rubber’. This was complemented by balloons, rubber bands, braces, boots for the army and other ideas that met with varying degrees of success.

In 1820 Thomas Hancock, an English manufacturer of rubber goods such as driving belts, industrial rollers and rubber hoses, invented a machine he called the ‘masticator’, which chewed up waste strips for re-use. It was discovered that the masticated rubber was more malleable, while maintaining much of its elasticity. In Scotland at the same time, Charles Macintosh had discovered a way of using rubber as waterproofing material, by a process he patented in 1823. Hancock and Macintosh joined forces in 1834, and three years later Hancock invented a machine for spreading rubber onto material.

Despite their beneficial qualities, such as waterproofing, rubber goods were still not particularly popular as they had some major flaws, including the fact that they dissolved marvellously. They also became pliant when warm and rigid when cold. Then in 1839 the American Charles Goodyear discovered that it was possible to stabilise rubber by mixing it with sulphur while exposing it to heat — a process he called vulcanisation — and the full versatility of this extraordinary substance became apparent.

Rubber goods could now be manufactured which had all the beneficial qualities of the material, such as durability, elasticity and variability, but which were not sticky, soluble or governed by the vagaries of the weather. The economic potential of rubber was now clearly evident. It played an important role in the Industrial Revolution, being employed in steam engines found in factories, mills, mines and railways. It made a triumphant entrance as a new and innovative material at the Great Exhibition of 1851, where shoes, airbeds, furniture and clothing made out of newly improved rubber were proudly displayed.

One of the most important rubber inventions was made in 1888, when an Irishman called John Boyd Dunlop produced the first pneumatic tyre. Solid rubber tyres had been used for the previous 18 years, but Dunlop’s new design, which he updated in 1890, immediately became popular. In 1895 Dunlop’s tyres were first used in motor cars, and with the mass production of cars just over the horizon the rubber industry had never looked healthier. The import levels of rubber over the nineteenth century bear witness to its irrepressible rise. In 1830 Britain had imported just 211 kg of crude rubber. This had risen to 10,000 kg in 1857, and by 1874 levels were just under six times as much again.

Questions 1–6

Do the following statements agree with the information given in Reading Passage 1?

In boxes 1–6 on your answer sheet, write:

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this

1. Rubber plants grow only in certain regions of the world.

2. Rubber was extracted in Mexico as early as the sixth century.

3. Rubber from the *Castilla elastica* plant is of poorer quality than that from the Para plant.

4. A French mathematician inspired real interest in rubber amongst Europeans.

5. The process of vulcanisation was discovered by accident.

6. Imports of crude rubber into Britain fell during the nineteenth century.

TRUE IELTS

Questions 7–13

Complete the summary below.

Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

Write your answers in boxes 7–13 on your answer sheet.

The Commercial Development of Rubber

Early European travellers gave accounts of various rubber objects in use in Central and South America, and these accounts created interest in the commercial exploitation of rubber. In 1818, **7.** _____ was produced using rubber, and in 1820 a machine was invented for recycling **8.** _____ of rubber. Over the next few years, other attempts were made to improve rubber, but some problems remained. For example, rubber products smelt bad when they were dissolved, and could turn either soft or **9.** _____ depending on the temperature. However, in 1839 a new process to **10.** _____ the substance greatly increased its potential. For example, rubber was used in the creation of the **11.** _____ industry during the Industrial Revolution. Then in 1888 the **12.** _____ was developed, and a few years later the **13.** _____ of the motor car began.

TRUE IELTS

ĐÁP ÁN:

1. FALSE – Ngay câu đầu tiên của đoạn 1 nêu rõ thực vật tạo ra cao su "phân bố khắp toàn cầu" (spread right across the globe) chứ không chỉ ở một số vùng nhất định.

Đoạn chứa đáp án:

The plants that produce rubber are spread right across the globe, and grow in many different habitats.

2. TRUE – Một bức tranh tường từ thế kỷ thứ 6 ở Mexico đã mô tả cảnh dâng lễ vật là cao su thô.

Đoạn chứa đáp án:

a wall painting dating back to the sixth century depicts a scene of a tributary offering of crude rubber.

3. NOT GIVEN – Văn bản có nhắc đến cả hai loại cây này nhưng không so sánh chất lượng của chúng.

Đoạn chứa đáp án:

Rubber was also used to make raincoats, shoes, jars, torches and musical instruments, all of which must have been made from the indigenous Castilla elastica, as the Para plant now favoured for rubber cultivation does not grow in the Mexican region.

4. TRUE – Charles de la Condamine (một nhà toán học Pháp) đã công bố lời kể về chuyến đi năm 1735, từ đó mới tạo ra sự quan tâm thực sự của người châu Âu.

Đoạn chứa đáp án:

However, no real interest in rubber was shown by any European until Charles de la Condamine, a French mathematician, published an account of his journey to South America in 1735.

5. NOT GIVEN – Văn bản nói Charles Goodyear đã khám phá ra quy trình này vào năm 1839 nhưng không đề cập việc đó là do tình cờ hay do nghiên cứu có chủ đích.

Đoạn chứa đáp án:

Then in 1839 the American Charles Goodyear discovered that it was possible to stabilise rubber by mixing it with sulphur while exposing it to heat - a process he called vulcanisation - and the full versatility of this extraordinary substance became apparent.

6.FALSE – Lượng nhập khẩu cao su vào Anh tăng mạnh trong thế kỷ 19 (từ 211 kg năm 1830 lên gần 60.000 kg vào năm 1874).

Đoạn chứa đáp án:

The import levels of rubber over the nineteenth century bear witness to its irrepressible rise. In 1830 Britain had imported just 211 kg of crude rubber. This had risen to 10,000kg in 1857, and by 1874 levels were just under six times as much again.

7. waterproof cloth – Năm 1818, James Syme lần đầu tiên dùng cao su để làm vải chống thấm nước.

Đoạn chứa đáp án:

In 1818 a British medical student named James Syme first used rubber to make waterproof cloth.

8. waste strips – Máy 'masticator' của Thomas Hancock được phát minh năm 1820 để tái chế các dải cao su phế thải.

Đoạn chứa đáp án:

In 1820 Thomas Hancock, an English manufacturer of rubber goods such as driving belts, industrial rollers and rubber hoses, invented a machine he called the 'piokle', which chewed up waste strips for re-use.

9. rigid – Cao su chưa lưu hóa sẽ trở nên mềm khi ấm và cứng (rigid) khi lạnh.

Đoạn chứa đáp án:

They also became pliant when warm and rigid when cold.

10. stabilise (hoặc **stabilize**) – Charles Goodyear tìm ra cách ổn định cao su bằng nhiệt và lưu huỳnh (lưu hóa).

Đoạn chứa đáp án:

Then in 1839 the American Charles Goodyear discovered that it was possible to stabilise rubber by mixing it with sulphur while exposing it to heat - a process he called vulcanisation - and the full versatility of this extraordinary substance became apparent.

11. steam engines – Cao su đóng vai trò quan trọng trong cuộc Cách mạng Công nghiệp (dùng cho động cơ hơi nước...).

Đoạn chứa đáp án:

It played an important role in the Industrial Revolution, being employed in the steam engines found in factories, mills, mines and railways.

12. pneumatic tyre – Năm 1888, John Boyd Dunlop đã sản xuất ra lốp bơm hơi đầu tiên.

Đoạn chứa đáp án:

One of the most important rubber inventions was made in 1888, when an Irishman called John Boyd Dunlop produced the first pneumatic tyre.

13. mass production – Vài năm sau đó, việc sản xuất hàng loạt xe hơi bắt đầu, thúc đẩy ngành cao su phát triển.

Đoạn chứa đáp án:

In 1895 Dunlop's tyres were first used in motor cars, and with the mass production of cars just over the horizon the rubber industry had never looked healthier.

READING PASSAGE 2

The fashion industry

A The fashion industry is a multibillion-dollar global enterprise devoted to the business of making and selling clothes. It encompasses all types of garments, from designer fashions to ordinary everyday clothing. Because data on the industry are typically reported for national economies and expressed in terms of its many separate sectors, total figures for world production of textiles and clothing are difficult to obtain. However, by any measure, the industry accounts for a significant share of world economic output.

B The fashion industry is a product of the modern age. Prior to the mid-19th century, virtually all clothing was handmade for individuals, either as home production or on order from dressmakers and tailors. By the beginning of the 20th century, with the development of new technologies such as the sewing machine, the factory system of production, and the growth of department stores and other retail outlets, clothing had increasingly come to be mass-produced in standard sizes and sold at fixed prices. Although the fashion industry developed first in Europe, today it is highly globalised, with garments often designed in one country, manufactured in another, and sold in a third. For example, an American fashion company might source fabric in China, have the clothes manufactured in Vietnam, finished in Italy, and shipped to a warehouse in the United States for distribution to retail outlets internationally.

C One of the first accomplishments of the Industrial Revolution in the 18th century was the partial automation of the spinning and weaving of wool, cotton, silk and other natural fibres. Today, these processes are highly automated and carried out by computer-controlled, high-speed machinery, and fabrics made from both natural fibres and synthetic fibres (such as nylon, acrylic and polyester) are produced. A growing interest in sustainable fashion (or 'eco-fashion') has led to greater use of environmentally friendly fibres, such as hemp. In addition, high-tech synthetic fabrics confer such properties as moisture absorption, stain resistance, retention or dissipation of body heat, and protection against fire, weapons, cold, ultraviolet radiation and other hazards. Fabrics are also produced with a wide range of visual effects through dyeing, weaving, printing and other processes. Together with fashion forecasters, fabric manufacturers work well in advance of the clothing production cycle to create fabrics with colours, textures and other qualities that anticipate consumer demand.

D Historically, very few fashion designers have become famous—brands such as Coco Chanel or Calvin Klein—who have been responsible for prestigious high-fashion collections. These designers are influential in the fashion world, but, contrary to popular belief, they do not dictate new fashions; rather, they endeavour to design clothes that will meet consumer demand. The vast majority of designers work anonymously for manufacturers, as part of design teams, adapting designs into marketable garments for average consumers. They draw inspiration from a wide range of sources, including film and television costumes, street clothing and active sportswear.

The fashion industry's traditional design methods, such as paper sketches and the draping of fabric on mannequins, have been supplemented or replaced by computer-assisted design

techniques. These allow designers to rapidly make changes to a proposed design and instantaneously share the proposed changes with colleagues—whether they are in the next room or on another continent.

E An important stage in garment production is the translation of the clothing design into templates, in a range of sizes, for cutting the cloth. Because the proportions of the human body change with increases or decreases in weight, templates cannot simply be scaled up or down. Template-making was traditionally a highly skilled profession. Today, despite innovations in computer programming, designs in larger sizes are difficult to adjust for every body shape. Whatever the size, the template—whether drawn on paper or programmed as a set of computer instructions—determines how fabric is cut into the pieces that will be joined to make a garment. For all but the most expensive clothing, fabric cutting is accomplished by computer-guided knives or high-intensity lasers that can cut many layers of fabric at once.

F The next stage of production is the assembly process. Some companies use their own production facilities for some or all of the manufacturing process, but the majority rely on separately owned manufacturing firms or contractors to produce garments to their specifications. In the field of women's clothing, manufacturers typically produce several product lines a year, which they deliver to retailers on predetermined dates. Technological innovation, including the development of computer-guided machinery, has resulted in the automation of some stages of assembly. Nevertheless, the fundamental process of sewing remains labour-intensive. In the late 20th century, China emerged as the world's largest producer of clothing because of its low labour costs and highly disciplined workforce.

Assembled items then go through various processes collectively known as 'finishing'. These include the addition of decorative elements, fasteners, brand-name labels and other labels (often legally required) specifying fibre content, laundry instructions and country of manufacture. Finished items are then pressed and packed for shipment.

G For much of the period following World War II, trade in textiles and garments was strictly regulated by purchasing countries, which imposed quotas and tariffs. Since the 1980s, these protectionist measures, which were intended (ultimately without success) to prevent textile and clothing production from moving from high-wage to low-wage countries, have gradually been abandoned. They have been replaced by a free-trade approach, under the regulatory control of global organisations. The advent of metal shipping containers and relatively inexpensive air freight has also made it possible for production to be closely tied to market conditions, even across globe-spanning distances.

Questions 14–20

*Reading Passage 2 has seven sections, **A–G**.*

Choose the correct heading for each section from the list of headings below.

*Write the correct number, **i-viii**, in boxes 14-20 on your answer sheet.*

List of Headings

i How new clothing styles are created

ii The rise of the fashion industry

iii Joining the garment pieces together

iv Producing materials with a range of features

v The importance of the fashion industry

vi Factors affecting international commerce

vii The attractions of becoming a fashion model

viii Making patterns for people with different figures

14 Section **A**

15 Section **B**

16 Section **C**

17 Section **D**

18 Section **E**

19 Section **F**

20 Section **G**

TRUE IELTS

Questions 21–24

Complete the summary below.

*Choose **NO MORE THAN TWO WORDS** from the passage for each answer.*

Write your answers in boxes 21–24 on your answer sheet.

The development of a modern fashion industry

Up until the middle of the 19th century, people generally wore handmade clothes. After that the situation changed, and by the 20th century many clothes were mass-produced. This development was partly due to inventions like the **21** _____. It was also the result of general changes in manufacturing systems, as well as the spread of shops like **22** _____. The changes also led to the standardisation of sizes and **23** _____. Today, despite the fact that the fashion industry originated in **24** _____, it has become a truly international enterprise.

Questions 25 and 26

*Choose **TWO** letters, **A–E**.*

Write the correct letters in boxes 25 and 26 on your answer sheet.

Which **TWO** of the following statements does the writer make about garment assembly?

A The majority of sewing is done by computer-operated machines.

B Highly skilled workers are the most important requirement.

C Most businesses use other companies to manufacture their products.

D Fasteners and labels are attached after the clothes have been made up.

E Manufacturers usually produce one range of women's clothing annually.

ĐÁP ÁN:

14. Section A: v (Tầm quan trọng của ngành công nghiệp thời trang - đóng góp đáng kể vào sản lượng kinh tế thế giới)

15. Section B: ii (Sự trỗi dậy của ngành công nghiệp thời trang - từ may tay sang sản xuất hàng loạt)

16. Section C: iv (Sản xuất vật liệu với nhiều tính năng - vải công nghệ cao, chống bẩn, chống cháy...)

17. Section D: i (Cách các phong cách quần áo mới được tạo ra - lấy cảm hứng từ phim ảnh, đường phố...)

18. Section E: viii (Tạo mẫu cho những người có vóc dáng khác nhau - dịch thiết kế sang các kích cỡ khác nhau)

19. Section F: iii (Gắn kết các mảnh may mặc lại với nhau - quá trình lắp ráp và hoàn thiện)

20. Section G: vi (Các yếu tố ảnh hưởng đến thương mại quốc tế - hạn ngạch, thuế quan và vận tải)

21. sewing machine (Sự phát triển này một phần là nhờ các phát minh như **máy may**)

22. department stores (Nó cũng là kết quả của sự lan rộng của các cửa hàng như **cửa hàng bách hóa**)

23. fixed prices (Các thay đổi cũng dẫn đến việc tiêu chuẩn hóa kích cỡ và **giá cố định**)

24. Europe (Mặc dù ngành thời trang bắt nguồn từ **Châu Âu**, nay nó đã trở thành doanh nghiệp quốc tế)

Questions 25 and 26: Multiple Choice (Choose TWO)

C (Hầu hết các doanh nghiệp sử dụng các công ty khác để sản xuất sản phẩm của họ - "the majority rely on separately owned manufacturing firms")

D (Các phụ kiện cài và nhãn mác được gắn sau khi quần áo đã được may xong - "Assembled items then go through... finishing... addition of... fasteners, brand-name labels")

TRUE IELTS

READING PASSAGE 3

Termite Mounds

Could the vast towers of mud constructed by insects in sub-Saharan Africa hold the key to our energy-efficient building of the future?

A

To most of us, termites are destructive insects which can cause damage on a devastating scale. But according to Dr Rupert Soar of Loughborough University's School of Mechanical and Manufacturing Engineering, these pests may serve a useful purpose for us after all. His multidisciplinary team of British and American engineers and biologists has set out to investigate the giant mounds built by termites in Namibia, in sub-Saharan Africa, as part of the most extensive study of these structures ever taken.

B

Termite mounds are impressive for their size alone; typically they are three metres high, and some as tall as eight metres have been found. They also reach far into the earth, where the insects 'mine' their building materials, carefully selecting each grain of sand they use. The termite's nest is contained in the central cavity of the mound, safely protected from the harsh environment outside. The mound itself is formed of an intricate lattice of tunnels, which split into smaller and smaller tunnels, much like a person's blood vessels.

C

This complex system of tunnels draws in air from the outside, capturing wind energy to drive it through the mound. It also serves to expel spent respiratory gases from the nest to prevent the termites from suffocating, so ensuring them a continuous provision of fresh, breathable air. So detailed is the design that the nest stays within three degrees of a constant temperature, despite variations on the outside of up to 50 °C, from blistering heat in the daytime to below freezing on the coldest nights. The mound also automatically regulates moisture in the air, by means of both its underground 'cellar', and evaporation from the top of the mound. Some colonies even had 'chimneys' at a height of 20 m to control moisture loss in the hottest regions of sub-Saharan Africa.

D

Furthermore, the termites have evolved in such a way as to outsource some of their biological functions. Part of their digestive process is carried out by a fungus, which they 'farm' inside the mound. This fungus, which is found nowhere else on earth, thrives in the constant and optimum environment of the mound. The termites feed the fungus with slightly chewed wood pulp, which the fungus then breaks down into a digestible sugary food to provide the insects with energy, and cellulose which they use for building. And, although the termites must generate waste, none ever leaves the structure, indicating that there is also some kind of internal waste-recycling system.

E

Scientists are so excited by the mounds that they have labelled them a 'super organism' because, in Soar's words, "They dance on the edge of what we would perceive to be cool or,

if you're too cold, you need to thrive: that's called homeostasis." What the termites have done is to move homeostatic function away from their body, into the structure in which they live. "As more information comes to light about the unique features of termite mounds, we may ultimately need to redefine our understanding of what constitutes a 'living' organism."

F

To reveal the structure of the mounds, Soar's team begins by filling and covering their plaster of Paris, a chalky white paste based on the mineral gypsum, which becomes rock-solid when dry. The researcher then carves the plaster of Paris into half-millimetre-thick slices, and photographs them sequentially. Once the pictures are digitally scanned, computer technology is able to recreate complex three-dimensional images of the mounds. These models have enabled the team to map termite architecture at a level of detail never before attained.

G

Soar hopes that the models will explain how termite mounds create a self-regulating living environment which manages to respond to changing internal and external conditions without drawing on any outside source of power. If they do, the findings could be invaluable in informing future architectural design, and could inspire buildings that are self-sufficient, environmentally friendly, and cheap to run. "As we approach a world of climate change and temperatures rise," he explains, "there will not be enough fuel to drive air conditioners around the world." It is hoped, says Soar, "that the findings will provide clues that aid the ultimate development of new kinds of human habitats, suitable for a variety of arid, hostile environments not only on the earth but maybe one day on the moon and beyond."

TRUE IELTS

*You should spend about 20 minutes on **Questions 27–40**, which are based on Reading Passage 3 below.*

Questions 27–33

*Reading Passage 3 has seven paragraphs **A–G**.*

Choose the correct heading for each paragraph from the list of headings below.

*Write the correct number, **i–ix**, in boxes 1–7 on your answer sheet.*

List of headings

i methods used to investigate termite mound formation
ii challenging our assumptions about the nature of life
iii reconsidering the termite's reputation
iv principal functions of the termite mound
v distribution of termite mounds in sub-Saharan Africa
vi some potential benefits of understanding termite architecture
vii the astonishing physical dimensions of the termite mound
viii termite mounds under threat from global climate change
ix a mutually beneficial relationship

27 Paragraph **A**

28 Paragraph **B**

29 Paragraph **C**

30 Paragraph **D**

31 Paragraph **E**

32 Paragraph **F**

33 Paragraph **G**

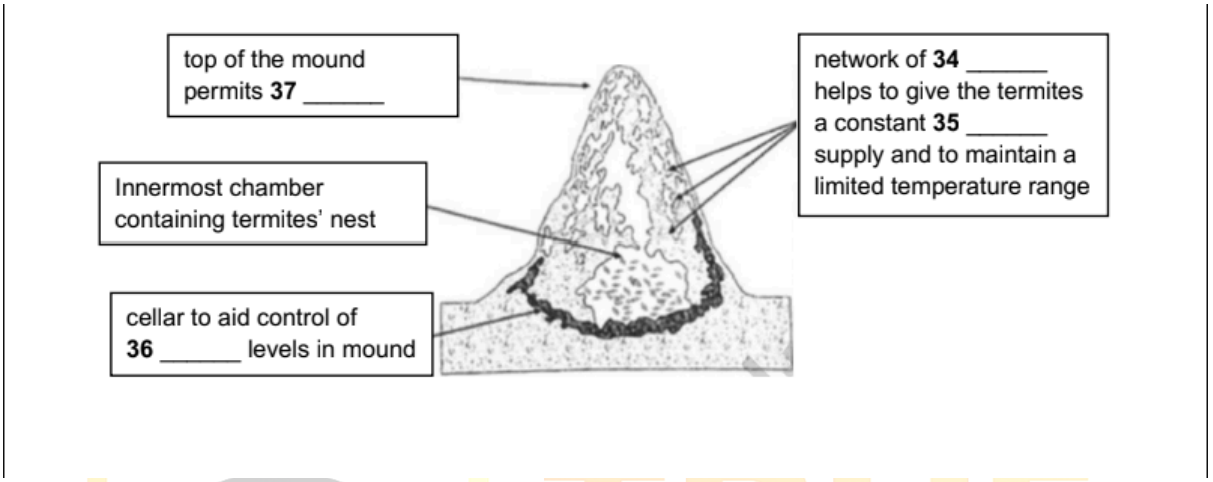
TRUE IELTS

Questions 34–37

Label the diagram below.

*Choose **ONE WORD ONLY** from the passage for each answer.*

Write your answers in boxes 34–37 on your answer sheet.



Questions 38–40

Do the following statements agree with the claims of the writer in Reading Passage 3?

In boxes 38–40 on your answer sheet, write

YES — if the statement agrees with the claims of the writer
NO — if the statement contradicts the claims of the writer
NOT GIVEN — if it is impossible to say what the writer thinks about this

38 The termite mound appears to process its refuse material internally.

39 Dr Soar's reconstruction involves scanning a single photograph of a complete mound into a computer.

40 New information about termite architecture could help people deal with future energy crises.

ĐÁP ÁN:

27. Paragraph A: iii (reconsidering the termite's reputation - xem xét lại danh tiếng của loài mối)

Đoạn chứa đáp án:

But according to Dr Rupert Soar of Loughborough University's School of Mechanical and Manufacturing Engineering, these pests may serve a useful purpose for us after all

28. Paragraph B: vii (the astonishing physical dimensions of the termite mound - kích thước vật lý kinh ngạc của gò mối)
Đoạn chứa đáp án:

Termite mounds are impressive for their size alone; typically they are three metres high, and some as tall as eight metres by found

29. Paragraph C: iv (principal functions of the termite mound - các chức năng chính của gò mối)
Đoạn chứa đáp án:

This complex system of tunnels draws in air from the outside, capturing wind energy to drive it through the mound. It also serves to expel spent respiratory gases from the nest to prevent the termites from suffocating, so ensuring them a continuous provision of fresh, breathable air.

30. Paragraph D: ix (a mutually beneficial relationship - một mối quan hệ đôi bên cùng có lợi)
Đoạn chứa đáp án:

Furthermore, the termites have evolved in such a way as to outsource some of their biological functions. Part of their digestive process in camera out by a fungus, which they 'farm' inside the mound. This fungus, which is found nowhere else on earth, thrives in the constant and optimum environment of the mound.

31. Paragraph E: ii (challenging our assumptions about the nature of life - thách thức các giả định của chúng ta về bản chất của sự sống)
Đoạn chứa đáp án:

‘As more information comes to light about the unique features of termite mounds, we may ultimately need to redefine our understanding of what constitutes a ‘living’ organism.

32. Paragraph F: i (methods used to investigate termite mound formation - các phương pháp được sử dụng để điều tra sự hình thành gò mối)
Đoạn chứa đáp án;

To reveal the structure of the mounds, Soar’s team begins by filling and covering their plaster of Paris, a chalky white paste based on the mineral gypsum, which becomes rock-solid when dry.

33. Paragraph G: vi (some potential benefits of understanding termite architecture - một số lợi ích tiềm năng từ việc hiểu kiến trúc của loài mối)
Đoạn chứa đáp án;

Soar hopes that the models will explain how termite mounds create a self-regulating living environment which manages to respond to changing internal and external conditions without drawing on any outside source of power. If they do, the findings could be invaluable in informing future architectural design, and could inspire buildings that are self-sufficient, environmentally, and cheap to run.

34. tunnels (mạng lưới đường hầm)
Đoạn chứa đáp án;

This complex system of tunnels draws in air from the outside, capturing wind energy to drive it through the mound. It also serves to expel spent respiratory gases from the nest to prevent the termites from suffocating, so ensuring them a continuous provision of fresh, breathable air.

35. air (cung cấp không khí liên tục)*Đoạn chứa đáp án;*

This complex system of tunnels draws in air from the outside, capturing wind energy to drive it through the mound. It also serves to expel spent respiratory gases from the nest to prevent the termites from suffocating, so ensuring them a continuous provision of fresh, breathable air.

36. moisture (hầm/cellar để kiểm soát mức độ ẩm)
Đoạn chứa đáp án:

The mound also automatically regulates moisture in the air, by means of best its underground ‘cellar’, and evaporation from the top of the mound.

37. evaporation (đỉnh gò cho phép sự bay hơi)
Đoạn chứa đáp án:

The mound also automatically regulates moisture in the air, by means of best its underground ‘cellar’, and evaporation from the top of the mound.

38. YES

> *Giải thích:* Đoạn D có nói: "none (waste) ever leaves the structure, indicating... internal waste-recycling system" (không có rác thải nào thoát ra ngoài, cho thấy hệ thống tái chế nội bộ).
> *Đoạn chứa đáp án:*

Part of their digestive process is carried out by a fungus, which they ‘farm’ inside the mound. This fungus, which is found nowhere else on earth, thrives in the constant and optimum environment of the mound.

39. NO

> *Giải thích:* Đoạn F mô tả quá trình cắt gò mối thành hàng ngàn lát mỏng và chụp ảnh tuần tự, chứ không phải quét "một tấm ảnh duy nhất" (single photograph).
> *Đoạn chứa đáp án:*

To reveal the structure of the mounds, Soar’s team begins by filling and covering their plaster of Paris, a chalky white paste based on the mineral gypsum, which becomes rock-solid when dry. The researcher's hen carves the plaster of Paris into half-millimatre-thick slices, and photograph them sequentially.

40. YES

Giải thích: Đoạn G khẳng định những phát hiện này có thể giúp xây dựng các tòa nhà tiết kiệm năng lượng khi đối mặt với biến đổi khí hậu và thiếu hụt nhiên liệu.
Đoạn chứa đáp án:

As we approach a world of climate change, we need temperatures to rise,' he explains. 'There will not be enough fuel to drive air conditioners around the world.' 'It is hoped,' says Soar, 'that the findings will provide clues that aid the ultimate development of new kinds of human habitats, suitable for a variety of arid, hostile environments not only on the earth but maybe one day on the moon and beyond.'

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